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WORKSHOP 1

REPORT

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EXECUTIVE SUMMARY

E-commerce, also known as electronic commerce, is the activity that people can buy or sell many types of products on online services or over the Internet. Everyone is allowed to browse, select, and purchase products from the comfort of their homes. This makes our life more convenient to get the things we want without stepping out from the room. In our knowledge, the online clothing store is the one of the electronic commerce that is commonly used by everyone in our life. However, there are three problem statements when operating a clothing store. First of all, inventory management may be inefficient such as update the real-time inventory data leading to the issues like overstocking or stockouts. It is also time consuming to manually handle many products information, manage orders and calculate the total amount of payment. Besides that, the buying experience of customer may be poor because of less product description make customers struggling in the colors, sizes or styles of the clothing. The objective of the online clothing store management system is to develop the online shopping system for users that can save their time without going to physical retail store. It also can ensure the accurate calculation for pricing of each order and price after different discount. It also will generate the report for analyzing total sales, customer behavior, inventory performance and gross profit. Furthermore, the online clothing store management system has 5 important module such as user management module, product management module, voucher and discount module, payment management module and report module. Lastly, there are three project significance which is to enhance experience of customers when purchase clothing using the online system. The system also wants to improve the efficiency of operation of the online clothing store and increase the growth of business.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In this digital age, e-commerce is the most popular activity in people's daily life. E-commerce which means electronic commerce is the activity that people can buy or sell many types of products on online services or over the Internet. E-commerce grows up unprecedented during the lockdown of 2020 which is the year of COVID-19 pandemic. Everyone is allowed to browse, select, and purchase products from the comfort of their homes. This makes our life more convenient to get the things we want to purchase without step out from the room. E-commerce has become an important tool for small and large businesses in the worldwide so the online clothing store is the one of the electronic commerce that commonly used by everyone in our life.

1.2 Problem Statement

1. Inventory management may be inefficient such as update the real-time inventory data leading to the issues like overstocking or stockouts.
2. Time consuming to manually handle many products information, manage orders and calculate the total amount of payment.
3. Buying experience of customer may be poor cause by less product description make customers struggling in the colors, sizes or styles of the clothing.

1.3 Objective(s) of the project

1. To develop the online shopping system for users that can save their time without going to physical retail store.
2. To ensure the accurate calculation for pricing of each order and price after different discount.
3. To generate the report for analyzing total sales, customer behavior, inventory performance and gross profit.

1.4 Scope

1.4.1 Modules to be developed

1. User Management Module

Allow customers and admins to login or sign up their account.

2. Product Management Module

Manage or view all the products for sale including products details, stock value, pricing and categories.

3. Discount Module

Able to apply discount to the orders.

4. Payment Management Module

Able to choose different payment methods.

5. Report Module

Provides data report about the sales.

1.4.2 Target User

1. Customers

Able to create or login an account to browse and purchase the products.

2. Admins

Able to manage the products and customer orders.

1.5 Project Significance

1. To enhance customers' experience when purchase clothing using the online system.
2. To improve the efficiency of operation of the online clothing store.
3. To increase the growth of business.

CHAPTER 2: ANALYSIS OF PROBLEM

2.1 Problem Decomposition Description

The clothing store always involves many challenges faced by both customers and staff managers when operating the business. Staffs need to track the inventory manually to update the daily stock which product is overstock or stockout. However, this will occur human error easily because have the large amounts of product.

The online clothing store management can help to solve the problem by update the stock levels and able to see the inventory at any times. Staff also need to calculate the amount of money for each order follow by the promotion activity. It will be complex when they have to calculate manually and prone to errors. In this situation, the system can help to calculate quickly and accurately so that will not have any mistakes of payment amount.

Besides, customers experience in physical store have to depends on the availability and knowledge of staff. Not all the experience of customers can get the best service quality of staff. However, in online clothing store, customers can check out the details of each product which have the size chart that customers can choose the most suitable size themselves without guidance of staffs.

2.2 Structure Chart

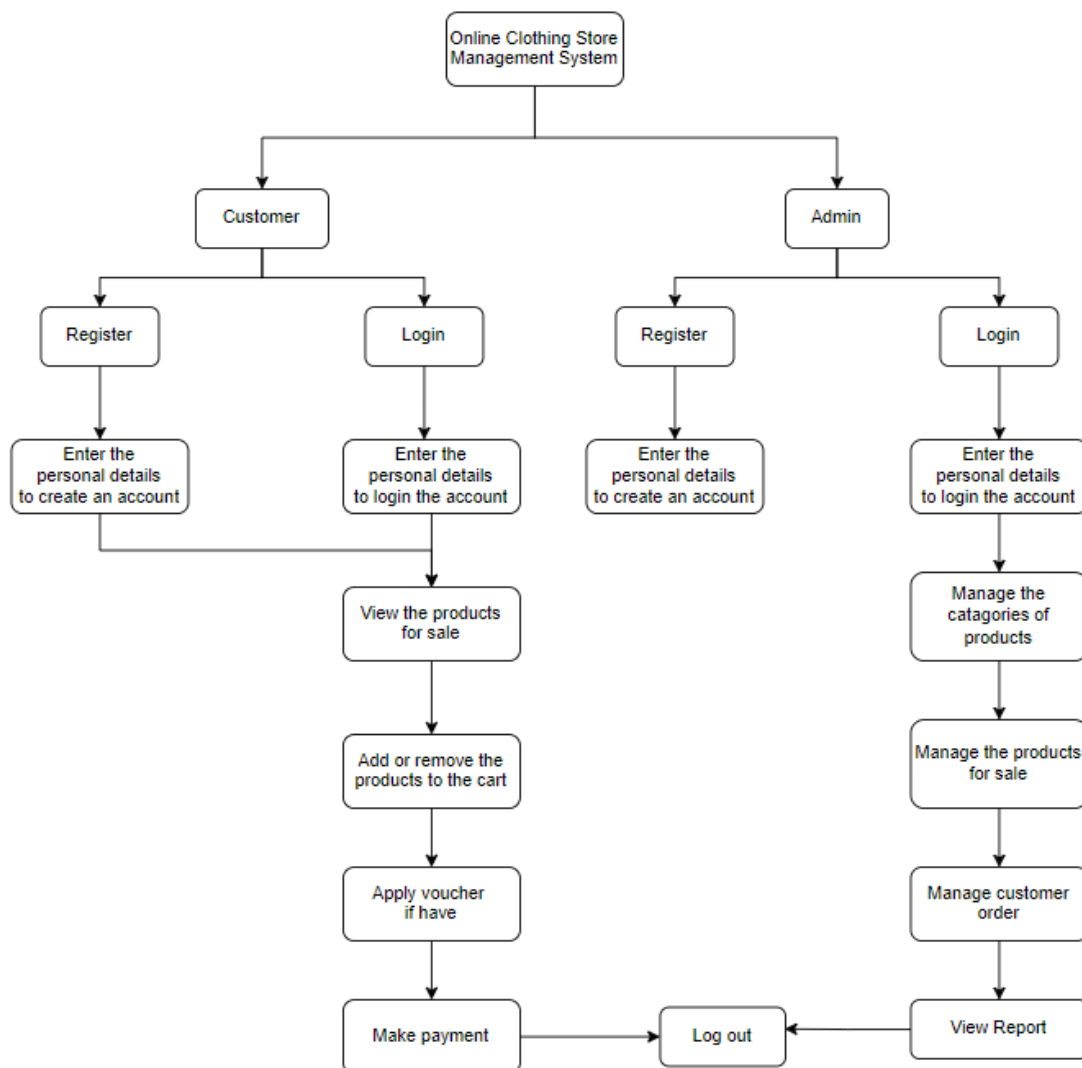


Figure 2.1: Structure Chart

CHAPTER 3: DESIGN

3.1 Flowchart

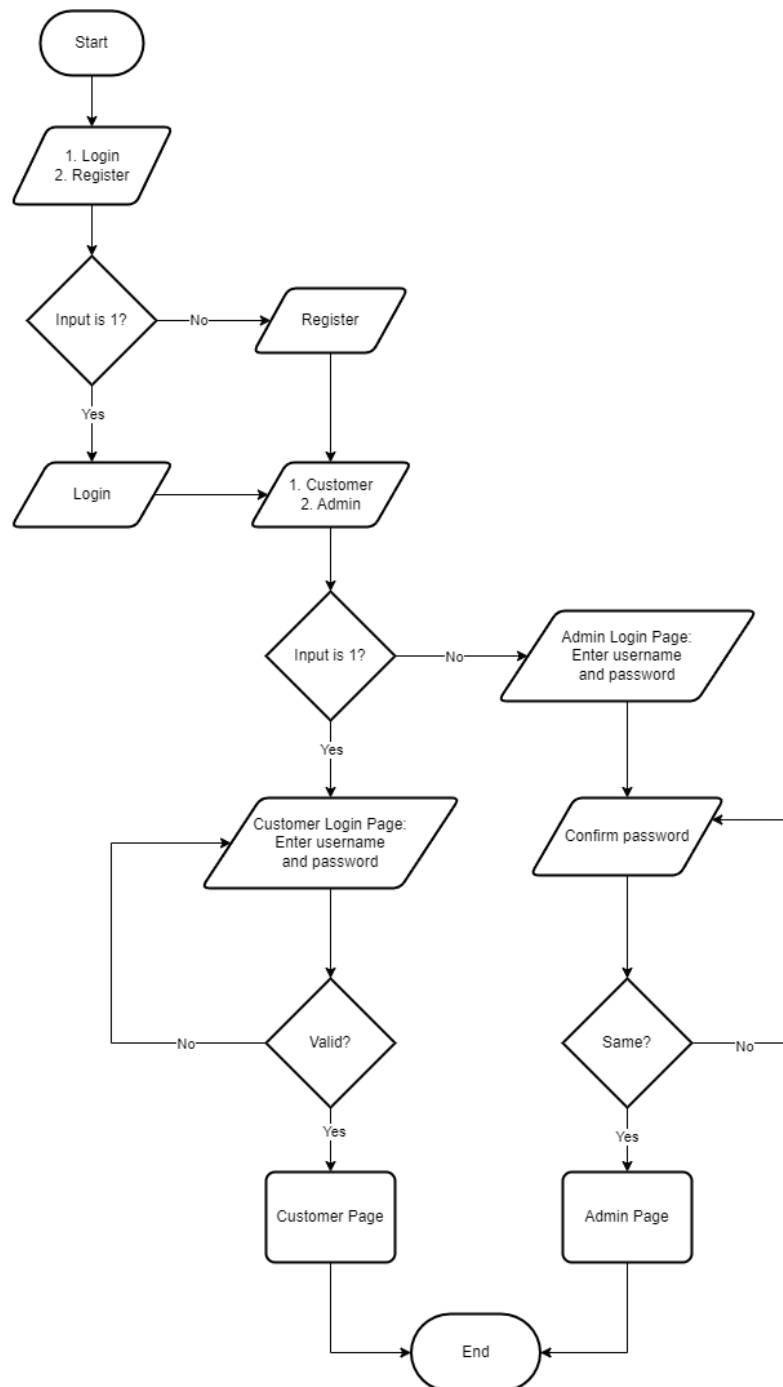


Figure 3.1: Login and Register

The users prompt to login or register, then is choose the role which is customer or admin when login or register. Users are required to input username and password. After that, the users will bring to admin page or customer page.

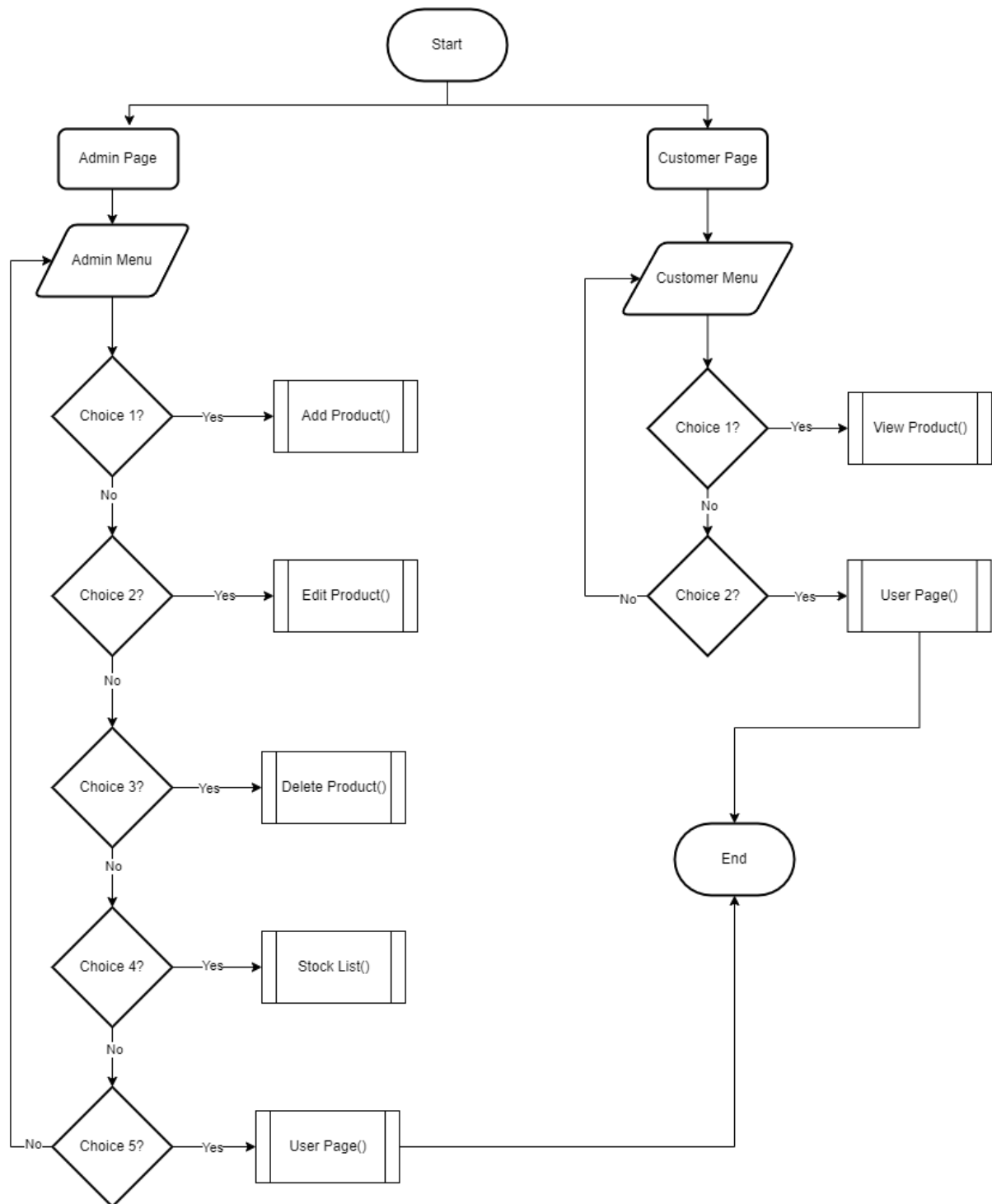


Figure 3.2: Product Management

For admin, they are prompt to select choice which are manage product (add product, edit product, delete product), and manage stock (view stock, update stock). While for customer, they are prompt to select choice which is view product.

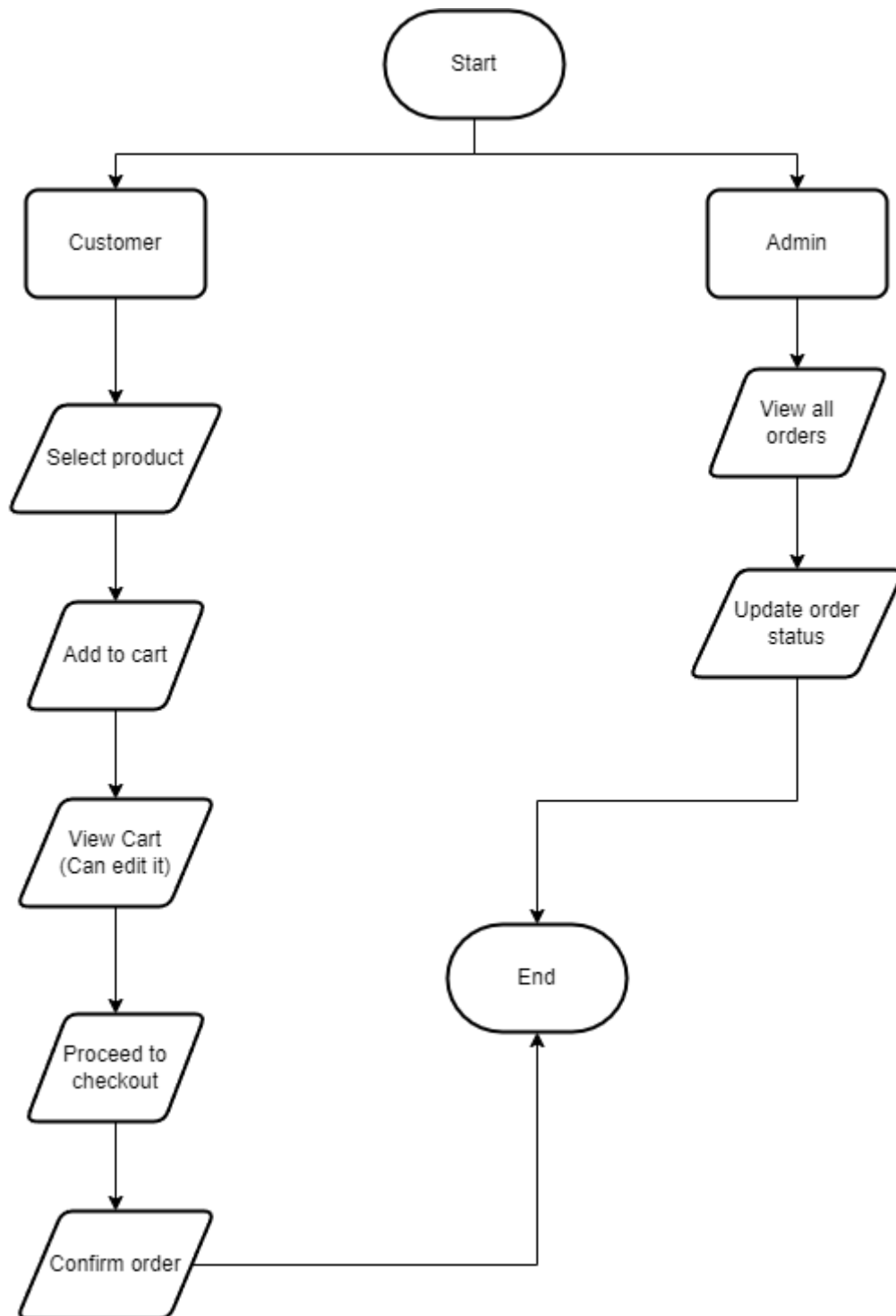


Figure 3.3: Order Management

For admin, they can manage order like view order and update order status.

For customer, they can select product and add to cart, view cart, update cart and delete items from cart. Then, they can proceed to checkout to confirm the order.

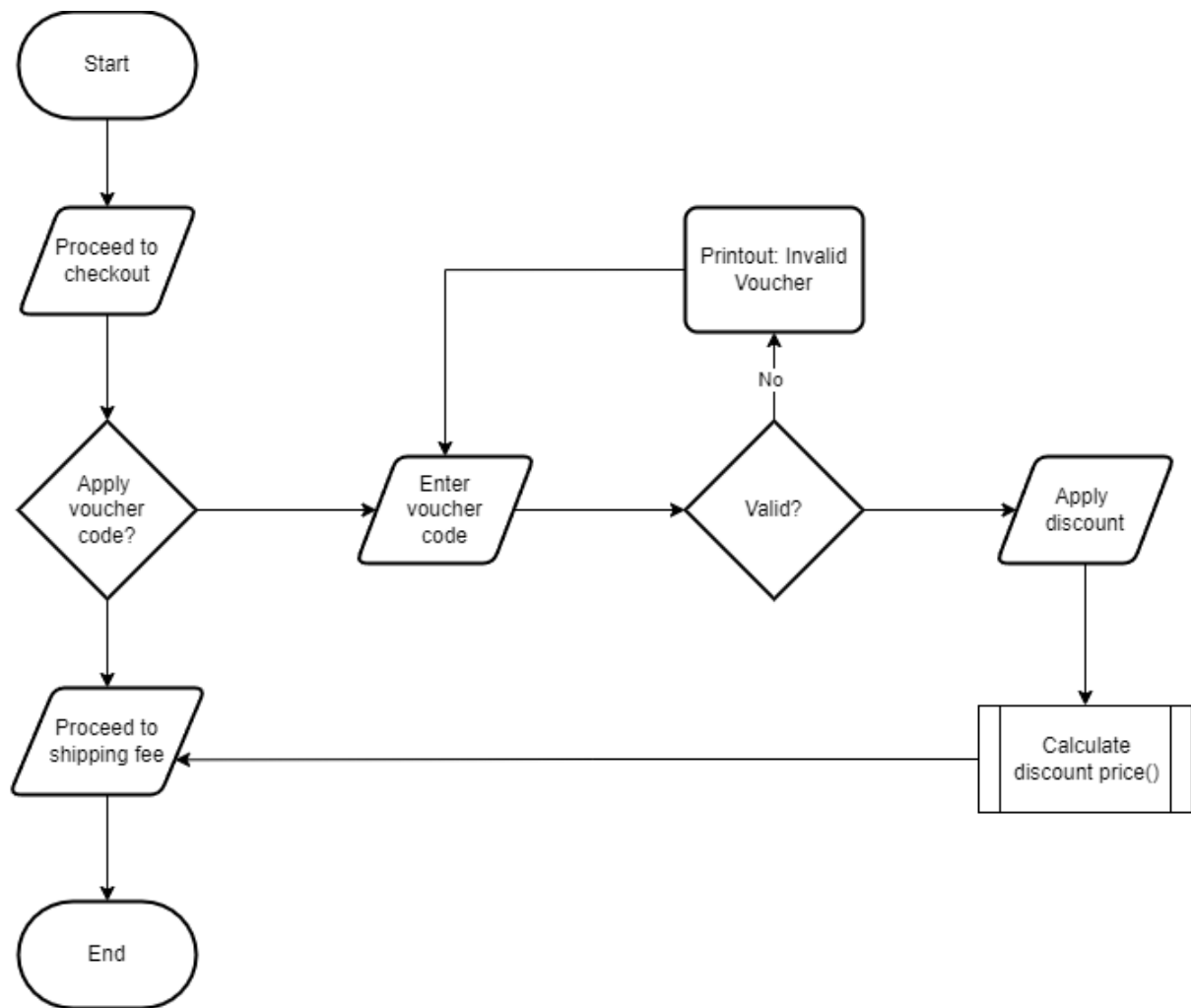


Figure 3.4: Voucher and Discount

When customer proceed to checkout, the system will apply fixed and tier discount first, then ask that whether want to apply voucher code.

After apply discount, the system calculates the discount price.

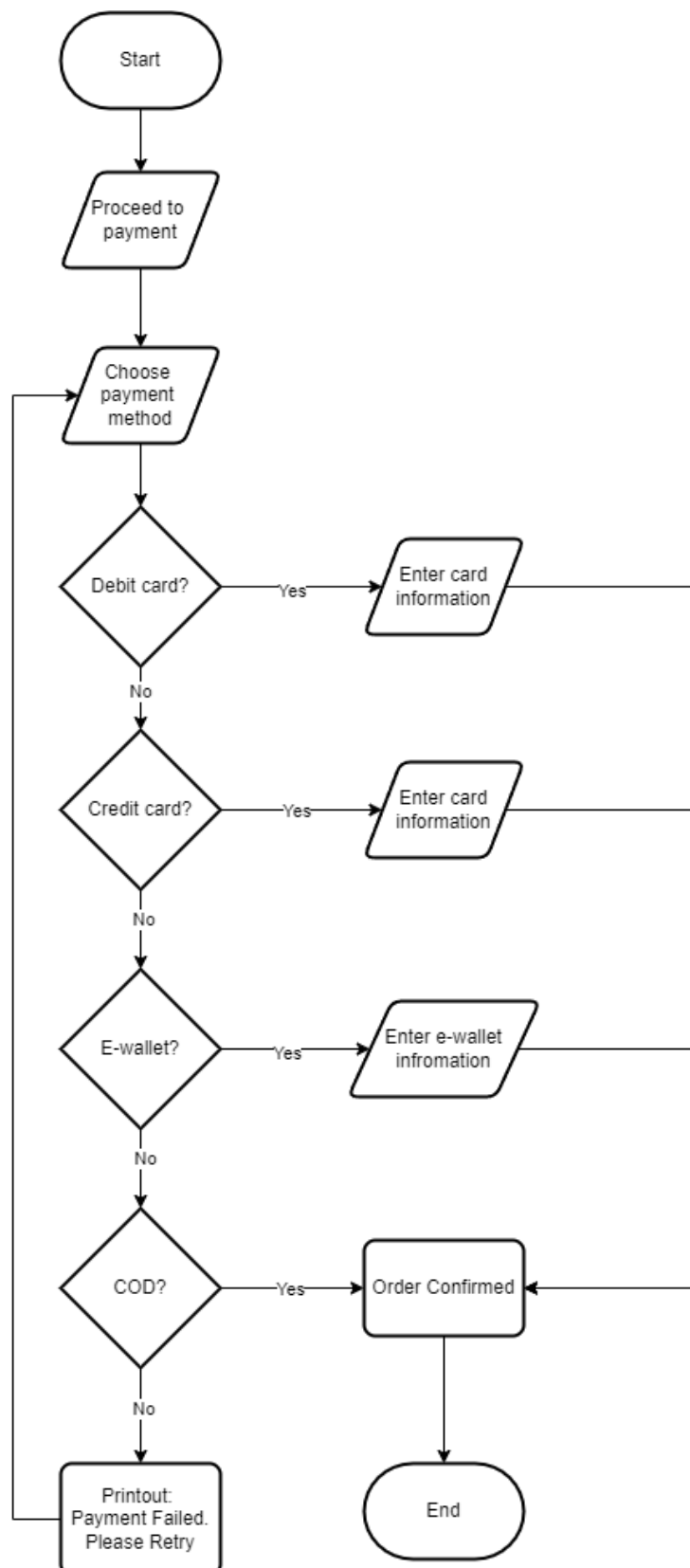


Figure 3.5: Payment

When customer proceed to do payment, they will be asked to choose the payment method. After payment done, the order confirmed and generate a receipt.

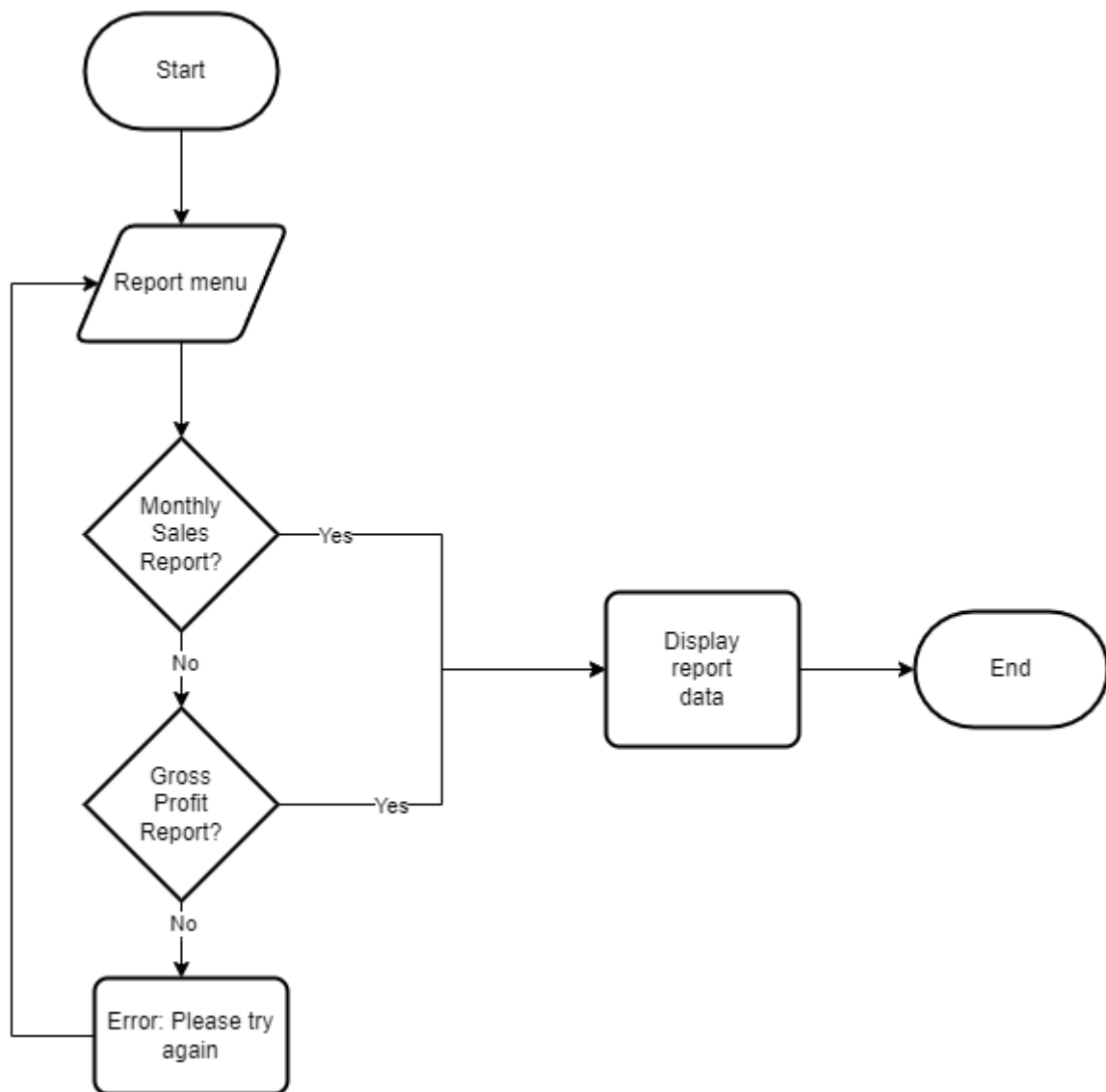


Figure 3.6: Report

Admin can generate report through system which are monthly sales report and gross profit report.

3.2 ERD

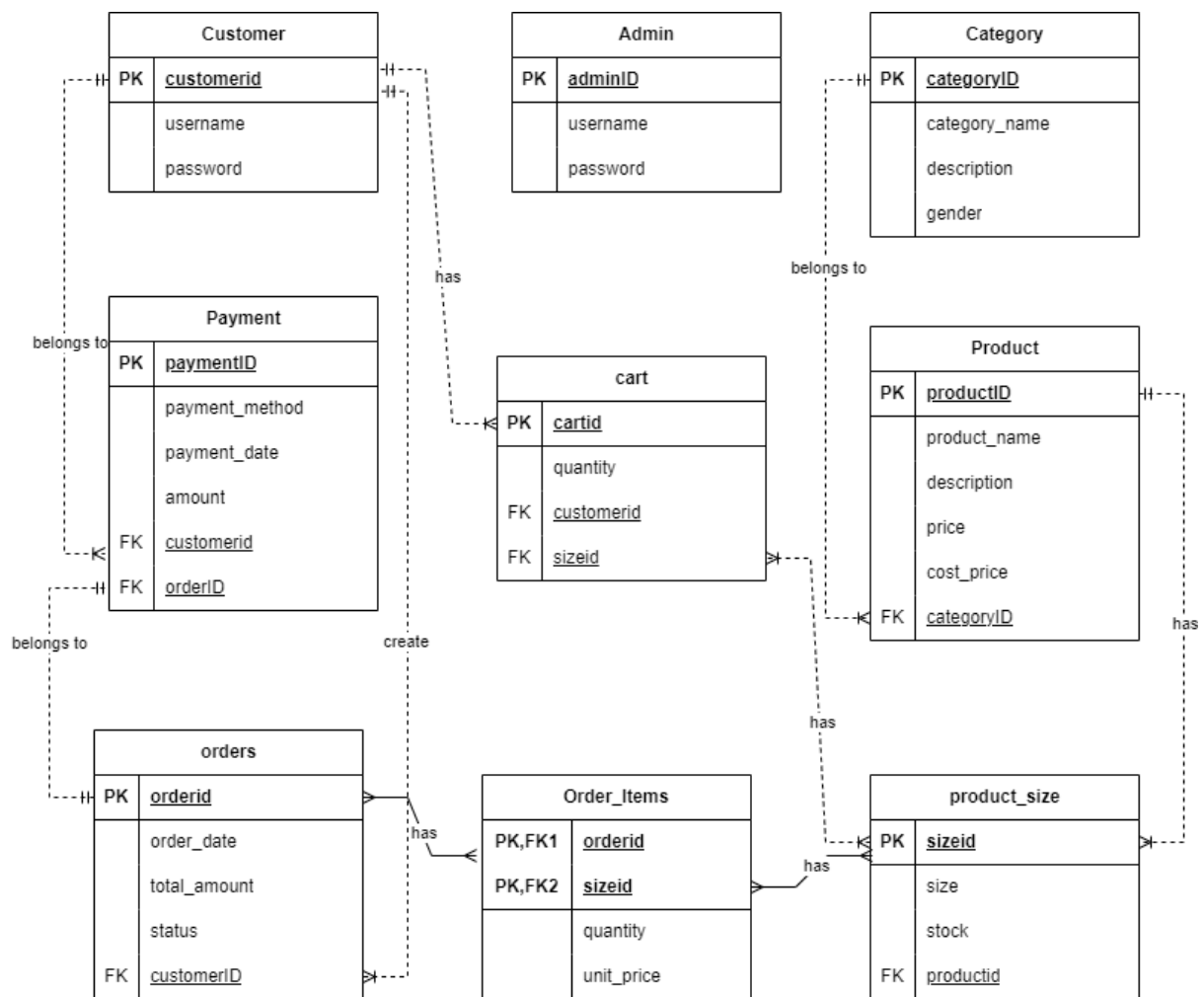


Figure 3.7: ERD

3.3 Data Dictionary

Table 3.1: CUSTOMER Table

CUSTOMER								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
customerid	varchar(5)	CXXXX		Yes		Yes	PK	
username	varchar(50)			Yes	NULL			
password	varchar(50)			Yes	NULL			

In customer table, customerid is PRIMARY KEY follow with the formatted id like C0001, C0002 and etc. The default value of username and password is NULL.

Table 3.2: ADMIN Table

ADMIN								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
adminid	varchar(5)	AXXXX		Yes		Yes	PK	
username	varchar(50)			Yes	NULL			
password	varchar(50)			Yes	NULL			

In admin table, adminid is the PRIMARY KEY follow with the formatted id like A0001, A0002 and etc. The default value of username and password is NULL.

Table 3.3: ORDERS Table

ORDERS								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
orderid	varchar(5)	OXXXX		Yes		Yes	PK	
customerid	varchar(5)	CXXXX		Yes		No	FK	CUSTOMER
order_date	datetime	YYYY-MM-DD HH:MM:SS		Yes				
total amount	decimal(10,2)	9999999999 . 99		Yes				
status	varchar(20)			Yes	Pending			

In orders table, orderid is the PRIMARY KEY follow with the formatted id like O0001, O0002 and etc. While customerid is the FOREIGN KEY refer to the customer table. The data type of order_date is datetime which is in format YYYY-MM-DD HH:MM:SS. The default value of status is “Pending”.

Table 3.4: ORDER_ITEM Table

ORDER_ITEM								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
orderid	varchar(6)	OXXXX		Yes		Yes	PK	
sizeid	varchar(5)	OXXXX		Yes		No	FK	ORDERS
quantity	varchar(4)	SXXX		Yes		No	FK	PRODUCT_SIZE
unit_price	int(11)			Yes				
	decimal(10,2)	9999999999 . 99		Yes				

In order_item table, orderitemid is the PRIMARY KEY follow with the formatted id like OI0001, OI0002 and etc. Orderid and sizeid are FOREIGN KEY refer to the orders table and product_size table.

Table 3.5: PRODUCT Table

PRODUCT								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
productid	varchar(5)	PXXXX		Yes		Yes	PK	
product_name	varchar(100)			Yes				
description	text			Yes				
price	decimal(10,2)	9999999999 . 99		Yes				
categoryid	varchar(4)			Yes		No	FK	CATEGORY
cost_price	decimal(10,2)	9999999999 . 99		Yes				

In product table, productid is the PRIMARY KEY follow with the formatted id like P0001, P0002 and etc. Categoryid is the FOREIGN KEY refer to the category table.

Table 3.6: PRODUCT_SIZE Table

PRODUCT_SIZE								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
sizeid	varchar(4)	SXXX		Yes		Yes	PK	
size	varchar(20)			Yes				
productid	varchar(5)	PXXXX		Yes		No	FK	PRODUCT
stock	int(11)			Yes				

In product_size table, sizeid is the PRIMARY KEY follow with the formatted id like S001, S002 and etc. Productid is the FOREIGN KEY refer to product table.

Table 3.7: CATEGORY Table

CATEGORY								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
categoryid	varchar(4)	CAXX		Yes		Yes	PK	
category_name	varchar(100)			Yes				
description	text			Yes				
gender	enum('Men', 'Women')			Yes				

In category table, categoryid is the PRIMARY KEY follow with the formatted id like CA01, CA02, and etc. Gender has two types which are “Men” and “Women”.

Table 3.8: CART Table

CART								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
cartid	varchar(5)	CTXXX		Yes		Yes	PK	
customerid	varchar(5)	CXXXX		Yes		No	FK	CUSTOMER
sizeid	varchar(4)	SXXX		Yes		No	FK	PRODUCT_SIZE
quantity	int(11)			Yes				

In cart table, cartid is the PRIMARY KEY follow with the formatted id like CT001, CT002, and etc. Customerid and sizeid is the FOREIGN KEY refer to customer table and product_size table.

Table 3.9: PAYMENT Table

PAYMENT								
ATTRIBUTE NAME	DATA TYPE AND SIZE	FORMAT	RANGE	REQUIRED?	DEFAULT VALUE	UNIQUE?	PK OR FK	FK REFERENCED TABLE
paymentid	varchar(6)	PMXXXX		Yes		Yes	PK	
orderid	varchar(5)	OXXXX		Yes		No	FK	ORDERS
customerid	varchar(5)	CXXXX		Yes		No	FK	CUSTOMER
payment_method	varchar(20)			Yes				
payment_date	datetime	YYYY-MM-DD HH:MM:SS		Yes				
amount	decimal(10,2)	9999999999 . 99		Yes				

In payment table, paymentid is the PRIMARY KEY follow with the formatted id like PM0001, PM0002, and etc. Orderid and customerid are FOREIGN KEY refer to orders table and customer table. The payment_date use datetime as data type in format YYYY-MM-DD HH:MM:SS.

3.4 Interface Design

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|=====|
|               Online Clothing Store                               |
|=====|

1. Register
2. Login
3. Exit
Enter your choice: |

```

Figure 3.8: Main menu

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|=====|
|               User Registration                                   |
|=====|

Enter a Username: Farah
Enter a Password: ***

Register as: 1. Customer 2. Admin
Enter your choice (1 or 2): 2

[INFO] Registration successful!
Your user ID is: A0006
Press any key to continue . . . |

```

Figure 3.9: Register new admin

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|=====|
|               User Registration                                   |
|=====|

Enter a Username: Alice
Enter a Password: ***

Register as: 1. Customer 2. Admin
Enter your choice (1 or 2): 1

[INFO] Registration successful!
Your user ID is: C0015
Press any key to continue . . . |

```

Figure 3.10: Register new customer


```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
=====
|                                     |
|                               User Login                           |
|                                     |
=====

Enter Username: Farah
Enter Password: ***
Login as: 1. Customer  2. Admin
Enter your choice (1 or 2): 2

[INFO] Login successful! Welcome, Farah!
Your user ID is: A0006
Press any key to continue . . . |

```

Figure 3.11: Admin login

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
=====
|                                     |
|                               User Login                           |
|                                     |
=====

Enter Username: Alice
Enter Password: ***
Login as: 1. Customer  2. Admin
Enter your choice (1 or 2): 1

[INFO] Login successful! Welcome, Alice!
Your user ID is: C0015
Press any key to continue . . . |

```

Figure 3.12: Customer login

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
=====
| Logged-in User ID: C0015                                           |
|                                     |
=====
|                               Customer Page                       |
|                                     |
=====

1. View Products
2. Add to Cart
3. View Cart
4. Return to Main Menu
Enter your choice: |

```

Figure 3.13: Customer menu

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|=====|=====|
| Logged-in User ID: C0015                                           |
|=====|=====|
|                                     |
|               Select Gender                                       |
|=====|=====|

```

Choose a gender:

1. Men
2. Women
3. Return to Main Menu

Enter your choice: |

Figure 3.14: Gender selection

After I choose choice one which is “View Product”, it jumps to this gender selection page.

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|=====|=====|
| Logged-in User ID: C0015                                           |
|=====|=====|
|                                     |
|               Select Category                                       |
|=====|=====|

```

Choose a category:

1. Casual Wear :
Casual attire is generally less formal than business attire. This type of clothes is stylish outfits for casual occasions and much more comfortable.
2. Everyday Attire :
A combination of comfort and individual flair, everyday attire is the foundation of daily fashion. Easy morning looks come from a basic dress or a pair of trusty denim pants worn with a favorite t-shirt.
3. Formal Clothing :
Formal wear is this type of clothing that is designed for working people or office wear. These clothes are tailored, fit well, and show professionalism and respect.
4. Sportswear :
Activewear designed for sports and physical activities. Manufactured of flexible and versatile fabric, this clothing offers excellent comfort.
5. Sleepwear :
Sleepwear is a particular type of clothing that is generally manufactured to wear during your bedtime is bedtime clothing. It's all about feeling relaxed and comfortable, aiding you to unwind after a long day.
6. Back to Gender Selection

Enter your choice: |

Figure 3.15: Category selection

This interface is the same for men or women.

Welcome to the Online Clothing Store ~Have a Nice Day~					
Logged-in User ID: C0015					
Products in Casual Wear					
Product ID	Name	Size	Stock	Price	Description
S003	Men's Casual T-shirt	Large	20	21.99	A comfortable cotton T-shirt for casual wear
S002	Men's Casual T-shirt	Medium	18	21.99	A comfortable cotton T-shirt for casual wear
S001	Men's Casual T-shirt	Small	47	21.99	A comfortable cotton T-shirt for casual wear
S033	Shoes	Medium	60	89.99	Comfortable leather shoes
S037	Shoes	Medium	60	89.99	Comfortable leather shoes
S041	Shoes	Medium	60	89.99	Comfortable leather shoes
S045	Shoes	Medium	60	89.99	Comfortable leather shoes
S049	Shoes	Medium	60	89.99	Comfortable leather shoes
S053	Shoes	Medium	60	89.99	Comfortable leather shoes
S032	Shoes	Small	50	89.99	Comfortable leather shoes
S036	Shoes	Small	50	89.99	Comfortable leather shoes
S040	Shoes	Small	50	89.99	Comfortable leather shoes
S044	Shoes	Small	50	89.99	Comfortable leather shoes
S048	Shoes	Small	50	89.99	Comfortable leather shoes
S052	Shoes	Small	50	89.99	Comfortable leather shoes
Would you like to add an item to your cart? (Y/N):					

Figure 3.16: Products in Casual Wear (Men)

The interface is same for women.

Welcome to the Online Clothing Store ~Have a Nice Day~					
Logged-in User ID: C0015					
Products in Everyday Attire					
Product ID	Name	Size	Stock	Price	Description
S007	Men's Jeans	32	54	39.99	Stylish denim jeans for everyday use
S008	Men's Jeans	34	70	39.99	Stylish denim jeans for everyday use
S009	Men's Jeans	36	30	39.99	Stylish denim jeans for everyday use
Would you like to add an item to your cart? (Y/N):					

Figure 3.17: Products in Everyday Attire (Men)

The interface is same for women.

Welcome to the Online Clothing Store ~Have a Nice Day~					
Logged-in User ID: C0015					
Products in Formal Clothing					
Product ID	Name	Size	Stock	Price	Description
S014	Men's Suit	L	15	129.99	Formal suit for business meetings
S013	Men's Suit	M	40	129.99	Formal suit for business meetings
S015	Men's Suit	XL	15	129.99	Formal suit for business meetings
Would you like to add an item to your cart? (Y/N):					

Figure 3.18: Products in Formal Clothing (Men)

The interface is same for women.

Welcome to the Online Clothing Store ~Have a Nice Day~					
Logged-in User ID: C0015					
Products in Sportswear					
Product ID	Name	Size	Stock	Price	Description
S019	Men's Running Shoes	7	56	59.99	Comfortable running shoes
S020	Men's Running Shoes	8	118	59.99	Comfortable running shoes
S021	Men's Running Shoes	9	56	59.99	Comfortable running shoes
Would you like to add an item to your cart? (Y/N):					

Figure 3.19: Products in Sportswear (Men)

The interface is same for women.

Welcome to the Online Clothing Store ~Have a Nice Day~					
Logged-in User ID: C0015					
Products in Sleepwear					
Product ID	Name	Size	Stock	Price	Description
S026	Men's Pajama Set	L	58	29.99	Comfortable sleepwear for men
S025	Men's Pajama Set	M	50	29.99	Comfortable sleepwear for men
S027	Men's Pajama Set	XL	40	29.99	Comfortable sleepwear for men
Would you like to add an item to your cart? (Y/N):					

Figure 3.20: Products in Sleepwear (Men)

The interface is same for women.

Welcome to the Online Clothing Store

~Have a Nice Day~

Logged-in User ID: C0015

Available Products

Product ID	Product Name	Size	Stock	Price	Description
S003	Men's Casual T-shirt	Large	20	21.99	A comfortable cotton T-shirt for casual wear
S002	Men's Casual T-shirt	Medium	18	21.99	A comfortable cotton T-shirt for casual wear
S001	Men's Casual T-shirt	Small	47	21.99	A comfortable cotton T-shirt for casual wear
S007	Men's Jeans	32	54	39.99	Stylish denim jeans for everyday use
S008	Men's Jeans	34	70	39.99	Stylish denim jeans for everyday use
S009	Men's Jeans	36	30	39.99	Stylish denim jeans for everyday use
S026	Men's Pajama Set	L	58	29.99	Comfortable sleepwear for men
S025	Men's Pajama Set	M	50	29.99	Comfortable sleepwear for men
S027	Men's Pajama Set	XL	40	29.99	Comfortable sleepwear for men
S019	Men's Running Shoes	7	56	59.99	Comfortable running shoes
S020	Men's Running Shoes	8	118	59.99	Comfortable running shoes
S021	Men's Running Shoes	9	56	59.99	Comfortable running shoes
S014	Men's Suit	L	15	129.99	Formal suit for business meetings
S013	Men's Suit	M	40	129.99	Formal suit for business meetings
S015	Men's Suit	XL	15	129.99	Formal suit for business meetings
S033	Shoes	Medium	60	89.99	Comfortable leather shoes
S037	Shoes	Medium	60	89.99	Comfortable leather shoes
S041	Shoes	Medium	60	89.99	Comfortable leather shoes
S045	Shoes	Medium	60	89.99	Comfortable leather shoes
S049	Shoes	Medium	60	89.99	Comfortable leather shoes
S053	Shoes	Medium	60	89.99	Comfortable leather shoes
S032	Shoes	Small	50	89.99	Comfortable leather shoes
S036	Shoes	Small	50	89.99	Comfortable leather shoes
S040	Shoes	Small	50	89.99	Comfortable leather shoes
S044	Shoes	Small	50	89.99	Comfortable leather shoes
S048	Shoes	Small	50	89.99	Comfortable leather shoes
S052	Shoes	Small	50	89.99	Comfortable leather shoes
S034	Socks	Large	80	9.99	Soft cotton socks
S038	Socks	Large	80	9.99	Soft cotton socks
S042	Socks	Large	80	9.99	Soft cotton socks
S046	Socks	Large	80	9.99	Soft cotton socks
S050	Socks	Large	80	9.99	Soft cotton socks
S054	Socks	Large	80	9.99	Soft cotton socks
S035	Socks	X-Large	100	9.99	Soft cotton socks
S039	Socks	X-Large	100	9.99	Soft cotton socks
S043	Socks	X-Large	100	9.99	Soft cotton socks
S047	Socks	X-Large	100	9.99	Soft cotton socks
S051	Socks	X-Large	100	9.99	Soft cotton socks
S055	Socks	X-Large	100	9.99	Soft cotton socks
S006	Women's Casual T-shirt	Large	40	19.99	A comfortable cotton T-shirt for casual wear
S005	Women's Casual T-shirt	Medium	90	19.99	A comfortable cotton T-shirt for casual wear
S004	Women's Casual T-shirt	Small	38	19.99	A comfortable cotton T-shirt for casual wear
S010	Women's Jeans	28	38	39.99	Stylish denim jeans for everyday use
S011	Women's Jeans	30	56	39.99	Stylish denim jeans for everyday use
S012	Women's Jeans	32	78	39.99	Stylish denim jeans for everyday use
S030	Women's Pajama Set	L	15	29.99	Comfortable sleepwear for women
S029	Women's Pajama Set	M	80	29.99	Comfortable sleepwear for women
S028	Women's Pajama Set	S	70	29.99	Comfortable sleepwear for women
S022	Women's Running Shoes	6	15	59.99	Comfortable running shoes
S023	Women's Running Shoes	7	90	59.99	Comfortable running shoes
S024	Women's Running Shoes	8	100	59.99	Comfortable running shoes
S017	Women's Suit	L	54	139.99	Formal suit for business meetings
S016	Women's Suit	M	48	139.99	Formal suit for business meetings
S018	Women's Suit	XL	39	139.99	Formal suit for business meetings
S031	q	s	30	12.00	q

Enter the Product ID of the product you wish to add to the cart (e.g., S001): |

Figure 3.21: Products List

The interface display out all the products and product's details let customer to choose to add to cart.

```

Enter the Product ID of the product you wish to add to the cart (e.g., S001): S020

Product Details:
Product ID: S020
Product Name: Men's Running Shoes
Size: 8
Stock Available: 118
Price: 59.99
Enter the quantity you wish to add to the cart: 2
[INFO] Item added to cart successfully.
Press any key to continue . . . |

```

Figure 3.22: Add product into the cart

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|                   ~Have a Nice Day~                             |
|                                     |
|-----|
| Logged-in User ID: C0015 |
|-----|
|               Your Cart               |
|-----|

Product ID   Product Name                Size   Quantity   Price   Total Price
-----
S020         Men's Running Shoes          8        2        59.99   119.98

Grand Total: RM119.98

Select an action:
1. Update Cart
2. Delete Item from Cart
3. Checkout
4. Return to Main Menu
Enter your choice (1/2/3/4): |

```

Figure 3.23: View cart

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|                   ~Have a Nice Day~                             |
|                                     |
|-----|
| Logged-in User ID: C0015 |
|-----|
|               Update Cart               |
|-----|

Product ID   Product Name                Size   Quantity   Price   Total Price
-----
S020         Men's Running Shoes          8        2        59.99   119.98

Enter the Product ID to update the quantity: S020
Enter the new quantity: 1
[INFO] Cart updated successfully.
Press any key to continue . . . |

```

Figure 3.24: Update cart

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|-----|
| Logged-in User ID: C0015                                           |
|-----|
|               Delete Cart                                          |
|-----|
| Product ID | Product Name | Size | Quantity | Price | Total Price |
|-----|-----|-----|-----|-----|-----|
| S020       | Men's Running Shoes | 8    | 1        | 59.99 | 59.99       |
| S010       | Women's Jeans      | 28   | 2        | 39.99 | 79.98       |
|-----|-----|-----|-----|-----|-----|
| Enter the Product ID to delete from the cart: S020
| [INFO] Item deleted successfully.
| Press any key to continue . . . |

```

Figure 3.25: Delete item from cart

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|-----|
| Logged-in User ID: C0015                                           |
|-----|
|               Checkout                                            |
|-----|
| Product ID | Product Name | Size | Quantity | Unit Price | Total Price |
|-----|-----|-----|-----|-----|-----|
| S010       | Women's Jeans | 28   | 5        | 39.99      | 199.95      |
|-----|-----|-----|-----|-----|-----|
| Grand Total Before Discounts: RM199.95
| [INFO] A fixed discount of RM10.00 is available.
| [INFO] A tiered discount of 5% (RM10.00) is available for spending over RM100.00.
| Do you have a coupon code? (Y/N): y
| Please enter your coupon code: SAVE15
| [INFO] Coupon code applied! You saved RM15.00.
|-----|
|
| Summary of Discounts:
|   Fixed Discount: RM10.00
|   Tier Discount: RM10.00
|   Coupon Discount: RM15.00
|
| Grand Total After Discounts: RM164.95
|
| Would you like to confirm checkout? (Y/N): y
|
| [INFO] Checkout completed successfully! Proceeding to payment...
| Press any key to continue . . . |

```

Figure 3.26: Checkout page

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|                   ~Have a Nice Day~                             |
|                                     |
|=====|
| Logged-in User ID: C0015                                           |
|=====|
|                               Process Payment                       |
|=====|

Order ID: 00056
Amount Due: RM164.95

Choose a payment method (e.g., Credit Card, PayPal, COD): cod

[INFO] You selected: cod

[INFO] Payment successful! Generating receipt...
Press any key to continue . . . |

```

Figure 3.27: Process payment

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|                   ~Have a Nice Day~                             |
|                                     |
|=====|
| Logged-in User ID: C0015                                           |
|=====|
|                               Payment Receipt                       |
|=====|

=====
|                                     |
|               PAYMENT RECEIPT                                     |
|=====|
Payment ID:      PM0056
Order ID:       00056
Customer ID:    C0015
Payment Method: cod
Payment Date:   2025-01-24 15:12:10
=====
Original Total:  RM199.95
Discount Applied: RM35.00
Amount Paid:     RM164.95
=====
|               Thank you for shopping with us!                   |
|               We look forward to serving you again!              |
|=====|
Press any key to continue . . . |

```

Figure 3.28: Payment receipt


```
=====
|                                     |
|           Welcome to the Online Clothing Store           |
|               ~Have a Nice Day~                          |
|                                     |
=====
Logged-in User ID: A0001
=====
                        Admin Page
=====

1. Manage Product
2. Manage Stock
3. Manage Order
4. Generate Report
5. Return to Main Menu
Enter your choice: |
```

Figure 3.29: Admin menu

```
=====
|                                     |
|           Welcome to the Online Clothing Store           |
|               ~Have a Nice Day~                          |
|                                     |
=====
Logged-in User ID: A0001
=====
                        Manage Product Menu
=====

1. Add Product
2. Edit Product
3. Delete Product
4. Back to Admin Menu
Choose an option: |
```

Figure 3.30: Manage product menu

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
=====
Logged-in User ID: A0001
=====
                                Add Products
=====

Available Categories:
Category ID   Category Name   Gender
-----
CA01         Casual Wear    Men
CA02         Casual Wear    Women
CA03         Everyday Attire  Men
CA04         Everyday Attire  Women
CA05         Formal Clothing  Men
CA06         Formal Clothing  Women
CA07         Sportswear      Men
CA08         Sportswear      Women
CA09         Sleepwear       Men
CA10         Sleepwear       Women

Enter the Category ID of the product: CA09
Enter product name: Cartoon pajamas
Enter product description: comfortable
Enter product price: 19.99
Enter product cost price: 10.99
Enter product size (e.g., S, M, L): m
Enter stock quantity for this size: 30
[INFO] Product added successfully!
Press any key to continue . . . |

```

Figure 3.31: Add product

Welcome to the Online Clothing Store

~Have a Nice Day~

Logged-in User ID: A0001

Edit Products

Available Products:

ProductID	CatID	Category Name	Product Name	Price	Cost	PriceDescription
P0001	CA01	Casual Wear	Men's Casual T-shirt	21.99	15.99	A comfortable cotton T-shirt for casual wear
P0012	CA01	Casual Wear	Black T-Shirt	19.99	10.99	Soft and Comfortable
P0014	CA01	Casual Wear	Shoes	89.99	60.99	Comfortable leather shoes
P0016	CA01	Casual Wear	Shoes	89.99	60.99	Comfortable leather shoes
P0018	CA01	Casual Wear	Shoes	89.99	69.99	Comfortable leather shoes
P0020	CA01	Casual Wear	Shoes	89.99	59.99	Comfortable leather shoes
P0022	CA01	Casual Wear	Shoes	89.99	79.99	Comfortable leather shoes
P0024	CA01	Casual Wear	Shoes	89.99	79.99	Comfortable leather shoes
P0002	CA02	Casual Wear	Women's Casual T-shirt	19.99	14.99	A comfortable cotton T-shirt for casual wear
P0015	CA02	Casual Wear	Socks	9.99	5.99	Soft cotton socks
P0017	CA02	Casual Wear	Socks	9.99	4.99	Soft cotton socks
P0019	CA02	Casual Wear	Socks	9.99	4.99	Soft cotton socks
P0021	CA02	Casual Wear	Socks	9.99	5.99	Soft cotton socks
P0023	CA02	Casual Wear	Socks	9.99	3.99	Soft cotton socks
P0025	CA02	Casual Wear	Socks	9.99	4.99	Soft cotton socks
P0003	CA03	Everyday Attire	Men's Jeans	39.99	34.99	Stylish denim jeans for everyday use
P0004	CA04	Everyday Attire	Women's Jeans	39.99	30.99	Stylish denim jeans for everyday use
P0005	CA05	Formal Clothing	Men's Suit	129.99	110.99	Formal suit for business meetings
P0006	CA06	Formal Clothing	Women's Suit	139.99	120.99	Formal suit for business meetings
P0007	CA07	Sportswear	Men's Running Shoes	59.99	49.99	Comfortable running shoes
P0008	CA08	Sportswear	Women's Running Shoes	59.99	50.99	Comfortable running shoes
P0009	CA09	Sleepwear	Men's Pajama Set	29.99	20.99	Comfortable sleepwear for men
P0026	CA09	Sleepwear	Cartoon pajamas	39.99	29.99	comfortable
P0010	CA10	Sleepwear	Women's Pajama Set	29.99	20.99	Comfortable sleepwear for women
P0013	CA10	Sleepwear	q	12.00	10.00	q

Enter the Product ID of the product to edit: P0026
 Enter new category ID (leave empty to keep current):
 Enter new product name (leave empty to keep current):
 Enter new description (leave empty to keep current):
 Enter new price (enter -1 to keep current): 32.99
 Enter new cost price (enter -1 to keep current): 15.99

Figure 3.32: Edit product

Welcome to the Online Clothing Store

~Have a Nice Day~

Logged-in User ID: A0001

Delete Products

Available Products:

ProductID	CatID	Category Name	Product Name	Price	Cost	PriceDescription
P0001	CA01	Casual Wear	Men's Casual T-shirt	21.99	15.99	A comfortable cotton T-shirt for casual wear
P0012	CA01	Casual Wear	Black T-Shirt	19.99	10.99	Soft and Comfortable
P0014	CA01	Casual Wear	Shoes	89.99	60.99	Comfortable leather shoes
P0016	CA01	Casual Wear	Shoes	89.99	60.99	Comfortable leather shoes
P0018	CA01	Casual Wear	Shoes	89.99	69.99	Comfortable leather shoes
P0020	CA01	Casual Wear	Shoes	89.99	59.99	Comfortable leather shoes
P0022	CA01	Casual Wear	Shoes	89.99	79.99	Comfortable leather shoes
P0024	CA01	Casual Wear	Shoes	89.99	79.99	Comfortable leather shoes
P0002	CA02	Casual Wear	Women's Casual T-shirt	19.99	14.99	A comfortable cotton T-shirt for casual wear
P0015	CA02	Casual Wear	Socks	9.99	5.99	Soft cotton socks
P0017	CA02	Casual Wear	Socks	9.99	4.99	Soft cotton socks
P0019	CA02	Casual Wear	Socks	9.99	4.99	Soft cotton socks
P0021	CA02	Casual Wear	Socks	9.99	5.99	Soft cotton socks
P0023	CA02	Casual Wear	Socks	9.99	3.99	Soft cotton socks
P0025	CA02	Casual Wear	Socks	9.99	4.99	Soft cotton socks
P0003	CA03	Everyday Attire	Men's Jeans	39.99	34.99	Stylish denim jeans for everyday use
P0004	CA04	Everyday Attire	Women's Jeans	39.99	30.99	Stylish denim jeans for everyday use
P0005	CA05	Formal Clothing	Men's Suit	129.99	110.99	Formal suit for business meetings
P0006	CA06	Formal Clothing	Women's Suit	139.99	120.99	Formal suit for business meetings
P0007	CA07	Sportswear	Men's Running Shoes	59.99	49.99	Comfortable running shoes
P0008	CA08	Sportswear	Women's Running Shoes	59.99	50.99	Comfortable running shoes
P0009	CA09	Sleepwear	Men's Pajama Set	29.99	20.99	Comfortable sleepwear for men
P0010	CA10	Sleepwear	Women's Pajama Set	29.99	20.99	Comfortable sleepwear for women
P0013	CA10	Sleepwear	q	12.00	10.00	q

Enter the Product ID of the product to delete: P0013
 Are you sure you want to delete product with ID P0013? (y/n): y

Figure 3.33: Delete product

Welcome to the Online Clothing Store
~Have a Nice Day~

Logged-in User ID: A0001

Manage Stock

1. View Stock

2. Update Stock

3. Return to Admin Menu

Choose an option: |

Figure 3.34: Manage stock menu

Welcome to the Online Clothing Store ~Have a Nice Day~			
Logged-in User ID: A0001			
View Stock			
ProductID	Product Name	Size	Stock
P0001	Men's Casual T-shirt	Small	47
P0001	Men's Casual T-shirt	Medium	18
P0001	Men's Casual T-shirt	Large	20
P0002	Women's Casual T-shirt	Small	38
P0002	Women's Casual T-shirt	Medium	90
P0002	Women's Casual T-shirt	Large	40
P0003	Men's Jeans	32	54
P0003	Men's Jeans	34	70
P0003	Men's Jeans	36	30
P0004	Women's Jeans	28	33
P0004	Women's Jeans	30	56
P0004	Women's Jeans	32	78
P0005	Men's Suit	M	40
P0005	Men's Suit	L	15
P0005	Men's Suit	XL	15
P0006	Women's Suit	M	48
P0006	Women's Suit	L	54
P0006	Women's Suit	XL	39
P0007	Men's Running Shoes	7	56
P0007	Men's Running Shoes	8	118
P0007	Men's Running Shoes	9	56
P0008	Women's Running Shoes	6	15
P0008	Women's Running Shoes	7	90
P0008	Women's Running Shoes	8	100
P0009	Men's Pajama Set	M	50
P0009	Men's Pajama Set	L	58
P0009	Men's Pajama Set	XL	40
P0010	Women's Pajama Set	S	70
P0010	Women's Pajama Set	M	80
P0010	Women's Pajama Set	L	15
P0013	q	s	30
P0014	Shoes	Small	50
P0014	Shoes	Medium	60
P0015	Socks	Large	80
P0015	Socks	X-Large	100
P0014	Shoes	Small	50
P0014	Shoes	Medium	60
P0015	Socks	Large	80
P0015	Socks	X-Large	100
P0014	Shoes	Small	50
P0014	Shoes	Medium	60
P0015	Socks	Large	80
P0015	Socks	X-Large	100
P0014	Shoes	Small	50
P0014	Shoes	Medium	60
P0015	Socks	Large	80
P0015	Socks	X-Large	100
P0014	Shoes	Small	50
P0014	Shoes	Medium	60
P0015	Socks	Large	80
P0015	Socks	X-Large	100
Press any key to continue . . .			

Figure 3.35: View stock

P0014	Shoes	Small	50
P0014	Shoes	Medium	60
P0015	Socks	Large	30
P0015	Socks	X-Large	100

Enter the Product ID of the product to update stock: P0014
Enter the Size of the product to update stock: Small
Enter the new stock quantity: 40|

Figure 3.36: Update stock

```
=====
|                                     |
|           Welcome to the Online Clothing Store           |
|               ~Have a Nice Day~                          |
|                                     |
|=====|
| Logged-in User ID: A0001 |
|=====|
|           Manage Order Menu           |
|=====|
```

1. View Orders
2. Update Order Status
3. Back to Admin Menu
Choose an option: |

Figure 3.37: Manage order menu

Welcome to the Online Clothing Store

~Have a Nice Day~

Logged-in User ID: A0001

View Orders

OrderID	Customer Name	Order Date	Status	Total Price
00001	Lili	2025-01-23 18:41:47	Delivered	79.98
00002	Jiayee	2025-01-24 00:58:26	Paid	323.96
00003	Lili	2025-01-24 01:08:02	Paid	43.98
00004	Jiayee	2025-01-24 03:34:08	Paid	219.96
00005	Jiayee	2025-01-24 03:41:18	Paid	361.95
00006	Jiayee	2025-01-24 03:44:57	Pending	199.96
00007	Jiayee	2025-01-24 03:50:38	Pending	139.96
00008	Jiayee	2025-01-24 03:56:20	Pending	199.96
00009	Jiayee	2025-01-24 03:58:16	Pending	79.97
00010	Jiayee	2025-01-24 03:59:31	Paid	179.96
00011	Jiayee	2025-01-24 04:07:04	Paid	199.96
00012	Jiayee	2025-01-24 04:10:59	Paid	309.96
00013	Jiayee	2025-01-24 07:06:24	Paid	339.97
00014	Lili	2025-01-24 07:17:21	Paid	279.94
00015	Lili	2025-01-24 07:31:12	Paid	219.97
00042	Jiayee	2025-01-24 09:14:12	Paid	259.98
00043	Ali	2025-01-24 09:47:53	Shipped	259.96
00044	Jiayee	2024-01-15 10:00:00	Paid	109.95
00045	Lili	2024-02-16 11:00:00	Paid	129.95
00046	Kathy	2024-03-17 12:00:00	Paid	139.94
00047	Kathy	2024-04-18 13:00:00	Paid	169.95
00048	Haha	2024-05-19 14:00:00	Paid	149.93
00049	Ali	2024-06-20 15:00:00	Paid	119.96
00050	Jiayee	2024-07-21 16:00:00	Paid	89.95
00051	Lili	2024-08-22 17:00:00	Paid	169.95
00052	Kathy	2024-09-23 18:00:00	Paid	89.95
00053	Kathy	2024-10-24 19:00:00	Paid	169.95
00056	Alice	2025-01-24 15:11:29	Paid	199.95

Press any key to continue . . . |

Figure 3.38: View Orders

00043	Ali	2025-01-24 09:47:53	Shipped
00049	Ali	2024-06-20 15:00:00	Paid
00055	Ali	2024-12-26 21:00:00	Paid
00056	Alice	2025-01-24 15:11:29	Paid
Enter the Order ID to update: 00056			
Current Order Status: Paid			
Enter new order status (e.g., 'Shipped', 'In Transit', 'Delivered', 'Pending', 'Paid'): Delivered			

Figure 3.39: Update Order

```

=====
|                                     |
|               Welcome to the Online Clothing Store               |
|               ~Have a Nice Day~                                   |
|                                     |
|-----|
| Logged-in User ID: A0001                                           |
|-----|
|               Generate Reports                                     |
|-----|
1. Monthly Sales Report
2. Gross Profit Report
3. Back to Admin Menu
Enter your choice: |

```

Figure 3.40: Generate report menu

Monthly Sales Report (After Discounts)

Month	Sales (RM)
2024-01	***** 779.97
2024-02	***** 789.97
2024-03	***** 699.98
2024-04	***** 1079.97
2024-05	***** 899.96
2024-06	***** 779.97
2024-07	***** 709.97
2024-08	***** 749.96
2024-09	***** 869.95
2024-10	***** 669.98
2024-11	***** 1239.97
2024-12	***** 979.95
2025-01	***** 3780.13

Press any key to continue . . . |

Figure 3.41: Monthly Sales Report

Month	Total Sales (RM)	Total Cost (RM)	Gross Profit (RM)
2024-01	959.94	151.98	807.96
2024-02	909.94	139.95	769.99
2024-03	899.96	161.98	737.98
2024-04	1209.94	194.95	1014.99
2024-05	1159.92	222.97	936.95
2024-06	929.94	133.96	795.98
2024-07	889.94	202.97	686.97
2024-08	919.92	194.95	724.97
2024-09	1069.90	202.97	866.93
2024-10	889.96	194.95	695.01
2024-11	1239.97	0.00	1239.97
2024-12	979.95	0.00	979.95
2025-01	6053.45	2282.59	3770.86

Gross Profit Report

Month	Sales (RM)
2024-01	***** 807.96
2024-02	***** 769.99
2024-03	***** 737.98
2024-04	***** 1014.99
2024-05	***** 936.95
2024-06	***** 795.98
2024-07	***** 686.97
2024-08	***** 724.97
2024-09	***** 866.93
2024-10	***** 695.01
2024-11	***** 1239.97
2024-12	***** 979.95
2025-01	***** 3770.86

Press any key to continue . . . |

Figure 3.42: Gross profit report

CHAPTER 4: IMPLEMENTATION

4.1 Naming Convention

```
// Admin Functions
static void ManageProduct(MYSQL* conn);
static void AddProduct(MYSQL* conn);
static void EditProduct(MYSQL* conn);
static void DeleteProduct(MYSQL* conn);
static void ManageStock(MYSQL* conn);
static void ViewStock(MYSQL* conn);
static void UpdateStock(MYSQL* conn);
static void ManageOrder(MYSQL* conn);
static void ViewOrder(MYSQL* conn);
static void UpdateOrderStatus(MYSQL* conn);
static void GenerateReportMenu(MYSQL* conn);

// Customer Functions
static void ViewProducts(MYSQL* conn);
static void AddToCart(MYSQL* conn);
static void ViewCart(MYSQL* conn);
static void UpdateCart(MYSQL* conn);
static void DeleteFromCart(MYSQL* conn);
static void Checkout(MYSQL* conn);
;
```

Figure 4.1: Function names

The figure above shows some example of function names being declared in the header file. The naming style used for most of the function names is camelCase. This makes it easier for users to read and understand.

```
void ProductManagement::AddProduct(MYSQL* conn) {
    string product_name, description, categoryid, size;
    double price, cost_price;
    int stock;
```

Figure 4.2: Variable declaration

The figure above shows an example of variable names being used in the add product function.

4.2 Function

```
class UserManagement {
private:
    static string currentUserId; // This will store the user_id of the logged-in user

public:
    static void DisplayHeader(const string& title);
    static string Login(MYSQL* conn);
    static void Register(MYSQL* conn);
    static string getCurrentUserId() { return currentUserId; }
    static void setCurrentUserId(const string& userId) { currentUserId = userId; }
};
```

```
class ProductManagement {
public:
    static void DisplayHeader(const string& title);
    static void DisplayAdminMenu();
    static void DisplayCustomerMenu();
    static void AdminPage(MYSQL* conn);
    static void CustomerPage(MYSQL* conn);

private:
    // Admin Functions
    static void ManageProduct(MYSQL* conn);
    static void AddProduct(MYSQL* conn);
    static void EditProduct(MYSQL* conn);
    static void DeleteProduct(MYSQL* conn);
    static void ManageStock(MYSQL* conn);
    static void ViewStock(MYSQL* conn);
    static void UpdateStock(MYSQL* conn);
    static void ManageOrder(MYSQL* conn);
    static void ViewOrder(MYSQL* conn);
    static void UpdateOrderStatus(MYSQL* conn);
    static void GenerateReportMenu(MYSQL* conn);

    // Customer Functions
    static void ViewProducts(MYSQL* conn);
    static void AddToCart(MYSQL* conn);
    static void ViewCart(MYSQL* conn);
    static void UpdateCart(MYSQL* conn);
    static void DeleteFromCart(MYSQL* conn);
    static void Checkout(MYSQL* conn);
};
```

```
class Payment {
public:
    static void DisplayHeader(const string& title);
    void ProcessPayment(MYSQL* conn, const string& userId, double grandTotal, const string& orderId);
    void GenerateReceipt(MYSQL* conn, const std::string& paymentId);
};
```

```

class Discount {
public:
    // Fixed discount amount
    static const double fixedDiscountAmount;

    // Tiered discount thresholds and percentages
    static const double tier1Threshold;
    static const double tier2Threshold;
    static const double tier1Discount;
    static const double tier2Discount;

    // Validate and calculate the coupon discount
    static double ApplyCouponCode(const string& couponCode);

    // Apply the fixed discount
    static double ApplyFixedDiscount(double& total);

    // Apply tiered discount
    static double ApplyTieredDiscount(double& total);
};

class Report {
public:
    void GenerateMonthlySalesReport(MYSQL* conn);
    void GenerateGrossProfitReport(MYSQL* conn);
private:
    void DisplayLineGraph(const std::map<std::string, double>& data, const std::string& title);
};

```

Figure 4.3: Functions

The figures above show all the existing functions in the system.

4.3 Array

```
// Reset gender count and array for each new query result
gender_count = 0;

system("cls");
DisplayHeader("Select Gender");
cout << "Choose a gender:\n";

// Loop through the query results and store them in the array
while ((row = mysql_fetch_row(res)) != nullptr && gender_count < MAX_GENDERS) {
    genders[gender_count] = row[0]; // Store gender in the array
    cout << gender_count + 1 << ". " << genders[gender_count] << endl;
    gender_count++;
}
cout << gender_count + 1 << ". Return to Main Menu\n";
cout << "Enter your choice: ";
cin >> gender_choice;
```

Figure 4.4: Array

This figure shows the code which has array. It retrieves gender data from the database and stores it in an array. After that, it displays the genders as a numbered list on the console that can allow user to choose a gender by entering the corresponding number. The `gender_count` variable tracks the number of genders fetched, and each gender is stored in the `genders` array. The customer can also choose to return to the main menu by selecting an option that corresponds to the number after the gender options.

4.4 Selection

```

void ProductManagement::AdminPage(MYSQL* conn) {
    DisplayHeader("Admin Page");
    int choice;
    do {
        DisplayAdminMenu();
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                ManageProduct(conn); // Function to manage products (add, edit, delete products)
                break;
            case 2:
                ManageStock(conn); // Function to manage stock levels (update stock)
                break;
            case 3:
                ManageOrder(conn); // Function to manage orders (view, update orders)
                break;
            case 4:
                GenerateReportMenu(conn); // Function to generate reports
                break;
            case 5:
                cout << "[INFO] Exiting admin menu...\n";
                return;
            default:
                cout << "[ERROR] Invalid choice. Please try again.\n";
                system("pause");
        }
    } while (true);
}

```

Figure 4.5: Switch-case

The figure shows the code with switch-case statement. It is used to execute different actions based on the choice of admin. It checks the value of choice and directs the program to the corresponding function. For instance, if the admin input 1, it calls `ManageProduct(conn)` to manage products. If 2 is chosen, it calls `ManageStock(conn)` to manage stock levels, and the others are the same action. If the admin enters an invalid option, the default case prompts an error message. The loop continues until the admin selects the option to exit (option 5).

4.5 Control

```

void ProductManagement::CustomerPage(MYSQL* conn) {
    DisplayHeader("Customer Page");
    int choice;
    do {
        DisplayCustomerMenu();
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                ViewProducts(conn);
                break;
            case 2:
                AddToCart(conn);
                break;
            case 3:
                ViewCart(conn);
                break;
            case 4:
                cout << "[INFO] Exiting customer menu...\n";
                return;
            default:
                cout << "[ERROR] Invalid choice. Try again." << endl;
                system("pause");
        }
    } while (true);
}

```

Figure 4.6: Do-while loop

The figure shows the code consists of do-while loop. It ensures that the customer menu is repeatedly displayed, allowing the user to choose an option until they decide to exit. It displays the customer menu and asks customer to input the choice. Based on the input, the program executes a specific case in the switch statement, such as view products or add items to the cart. If the customer enters an invalid choice, the default case prompts an error message., and the loop continues. The loop only stops when the customer selects option 4 to exit.

```

while (!validMethod) {
    cout << "\nChoose a payment method (e.g., Credit Card, PayPal, COD): ";
    cin.ignore();
    getline(cin, paymentMethod);

    // Convert to lowercase for case-insensitive comparison
    string lowerMethod = paymentMethod;
    transform(lowerMethod.begin(), lowerMethod.end(), lowerMethod.begin(), ::tolower);

    // Check for valid payment methods
    if (lowerMethod == "credit card" || lowerMethod == "paypal" || lowerMethod == "cod") {
        validMethod = true; // Input is valid
    }
    else {
        cout << "[ERROR] Invalid payment method! Please choose from Credit Card, PayPal, or COD.\n";
    }
}

```

Figure 4.7: While loop

The figure shows the code snippet has while loop that continuously prompts the customer to choose a payment method until a valid method is entered. Whatever the customer input, it will be converted to lowercase to allow case-insensitive comparison. The loop checks if the entered method matches one of the valid options (Credit Card, PayPal, or COD). If the input is valid, validMethod is set to true, exiting the loop. If the input is invalid, an error message is shown, and the loop prompts the customer input again.

4.6 Pointer

```
// Dynamically allocate memory for the payment method
string* paymentMethod = new string;
bool validMethod = false; // Flag to check for valid input

while (!validMethod) {
    cout << "\nChoose a payment method (e.g., Credit Card, PayPal, COD): ";
    cin.ignore();
    getline(cin, *paymentMethod); // Use the pointer dereferencing to read input

    // Convert to lowercase for case-insensitive comparison
    string lowerMethod = *paymentMethod;
    transform(lowerMethod.begin(), lowerMethod.end(), lowerMethod.begin(), ::tolower);

    // Check for valid payment methods
    if (lowerMethod == "credit card" || lowerMethod == "paypal" || lowerMethod == "cod") {
        validMethod = true; // Input is valid
    }
    else {
        cout << "[ERROR] Invalid payment method! Please choose from Credit Card, PayPal, or COD.\n";
    }
}
```

Figure 4.8: Pointer

The figure shows the code snippet use pointer concept. The pointer is `paymentMethod`, declared as `string* paymentMethod = new string;`, which dynamically allocates memory for a string. It is used to store the user's input via `getline(cin, *paymentMethod)` and manipulated using `*paymentMethod` to access the actual string value. This pointer-based approach is unnecessary for this use case, as a regular string variable would suffice.

4.7 Error Handling

```
// Ask customer to input quantity
cout << "Enter the quantity you wish to add to the cart: ";
cin >> quantity;

if (quantity <= 0) {
    cout << "[ERROR] Quantity must be greater than zero.\n";
    system("pause");
    return;
}

if (quantity > available_stock) {
    cout << "[ERROR] Insufficient stock. Available stock: " << available_stock << endl;
    system("pause");
    return;
}
```

Figure 4.9: Error handling for validation quantity

The figure shows error handling in the code snippet that ensures the customer inputs a valid quantity for adding to the cart. If the quantity input is less than or equal to zero and displays an error message to prevent invalid input. It also checks if the requested quantity exceeds the available stock and displays an error message with the available stock, ensure the customer cannot order more than what's in stock. In both cases, the system pauses for the user to read the message and then exits the function to prevent further processing.

```

// Function to register users
void UserManagement::Register(MYSQL* conn) {
    DisplayHeader("User Registration");
    string username, password, userType;

    cout << "Enter a Username: ";
    getline(cin, username);

    // Validate username input
    if (username.empty()) {
        cout << "[ERROR] Username cannot be empty." << endl;
        system("pause");
        return;
    }
}

```

Figure 4.10: Error handling for user registration

This figure shows error handling in the code which ensures that a username must be provided during user registration. If the input for the username is empty, it displays an error message, pauses to let the user read it, and exits the function. This prevents proceeding with invalid or incomplete registration data, ensuring the integrity of the user input.

CHAPTER 5: CONCLUSION

5.1 Constraints

One of the limitations of this system is that it has less detailed product information due to the lack of images. Without images, customers will struggle to visualize how the product looks like just by reading the description. This will lead to potential dissatisfaction and wasted money if the purchased items do not meet their expectations. Products may also appear different in person than described from what they imagined.

Other than that, new buyers cannot assess product quality as well due to the absence of reviews and ratings from previous buyers. It will be hard for them to make informed decisions and increases the risk of purchasing poor-quality items. This will lead to disappointment and a negative shopping experience for new customers.

5.2 Future Improvements

To improve and address the system's current limitations, incorporating images for a comprehensive and accurate product description is highly recommended. Visual representations play a crucial role in helping customers better understand the appearance, features, and details of the products they intend to purchase. This addition can significantly reduce the chances of customer dissatisfaction, as buyers will have a clearer idea of what they are purchasing, thereby minimizing misunderstandings or mismatched expectations.

Furthermore, the inclusion of reviews and ratings from previous customers should be prioritized. Customer feedback provides valuable insights into the quality, functionality, and overall performance of the products. Potential buyers can rely on these reviews to make more informed decisions, fostering a sense of trust and transparency within the shopping experience.

By implementing these features, the system can enhance the overall user experience and create a more engaging and reliable platform. Improved satisfaction levels will likely result in stronger customer loyalty, repeat purchases, and a more positive perception of the online shopping experience, ultimately benefiting both the customers and the business.

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